FINANCIAL INNOVATION

IMPLICATIONS FOR PAYMENTS AND POLICY

FEDERAL RESERVE BANK OF KANSAS CITY
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FEDERAL RESERVE BANK OF KANSAS CITY

Financial innovation

Gorton (2009), “The Subprime Panic”, Jackson Hole Symposium

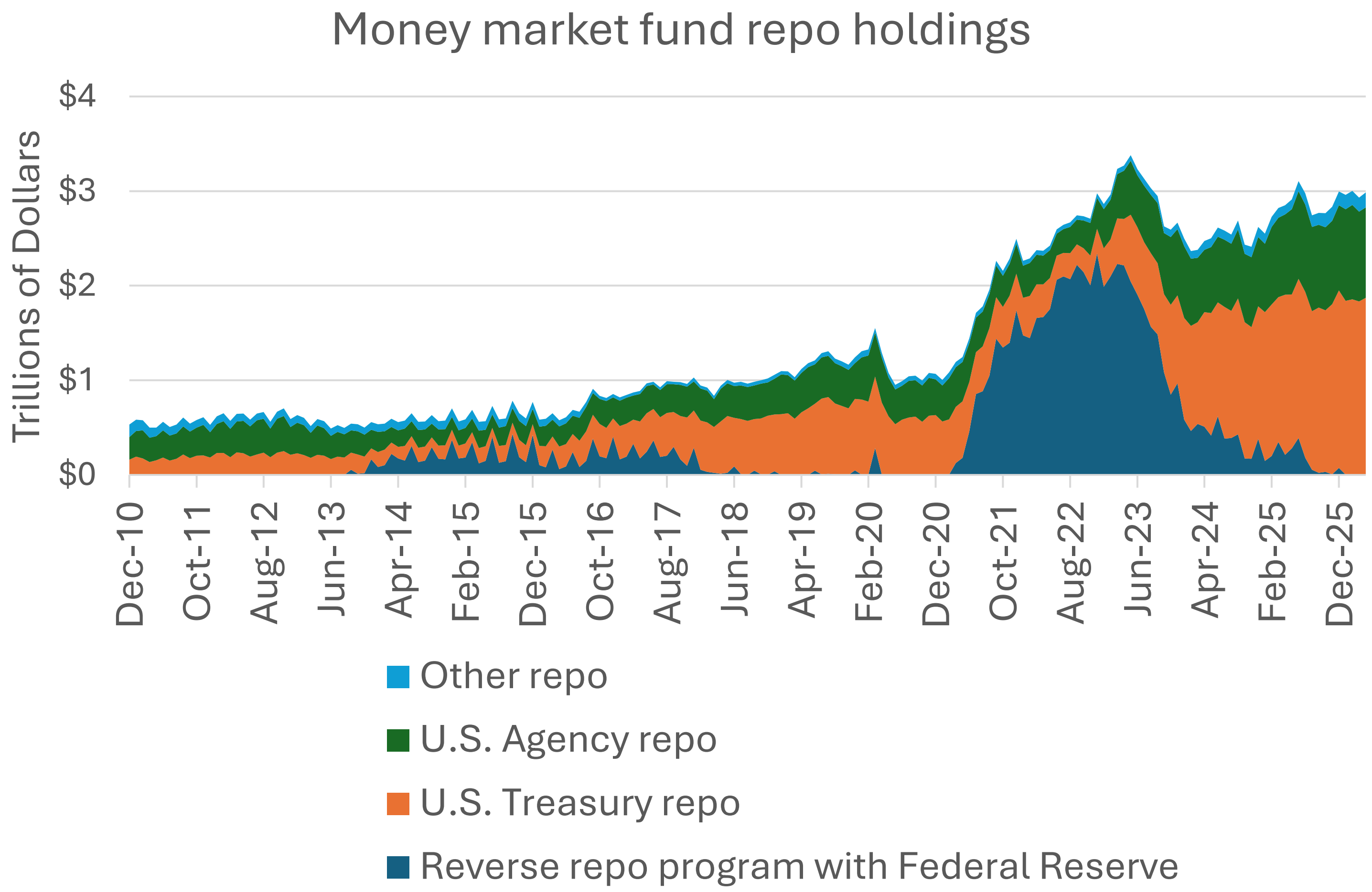
2000-2008: Rise of shadow banking, securitization, repo, asset-backed CP

- Repo, ABCP as the “money” deposit in a shadow bank
- Supporting mortgage lending
- Non-bank security dealers

GFC

- Roughly $2 trillion of this type of money
- Run on repo and ABCP in the financial crisis

Growth of repo, post-GFC



Estimate of TOTAL repo market ~ $12 trillion today

Hempel, Kahn, and Shephard (2025)

New money markets

For cash investors:

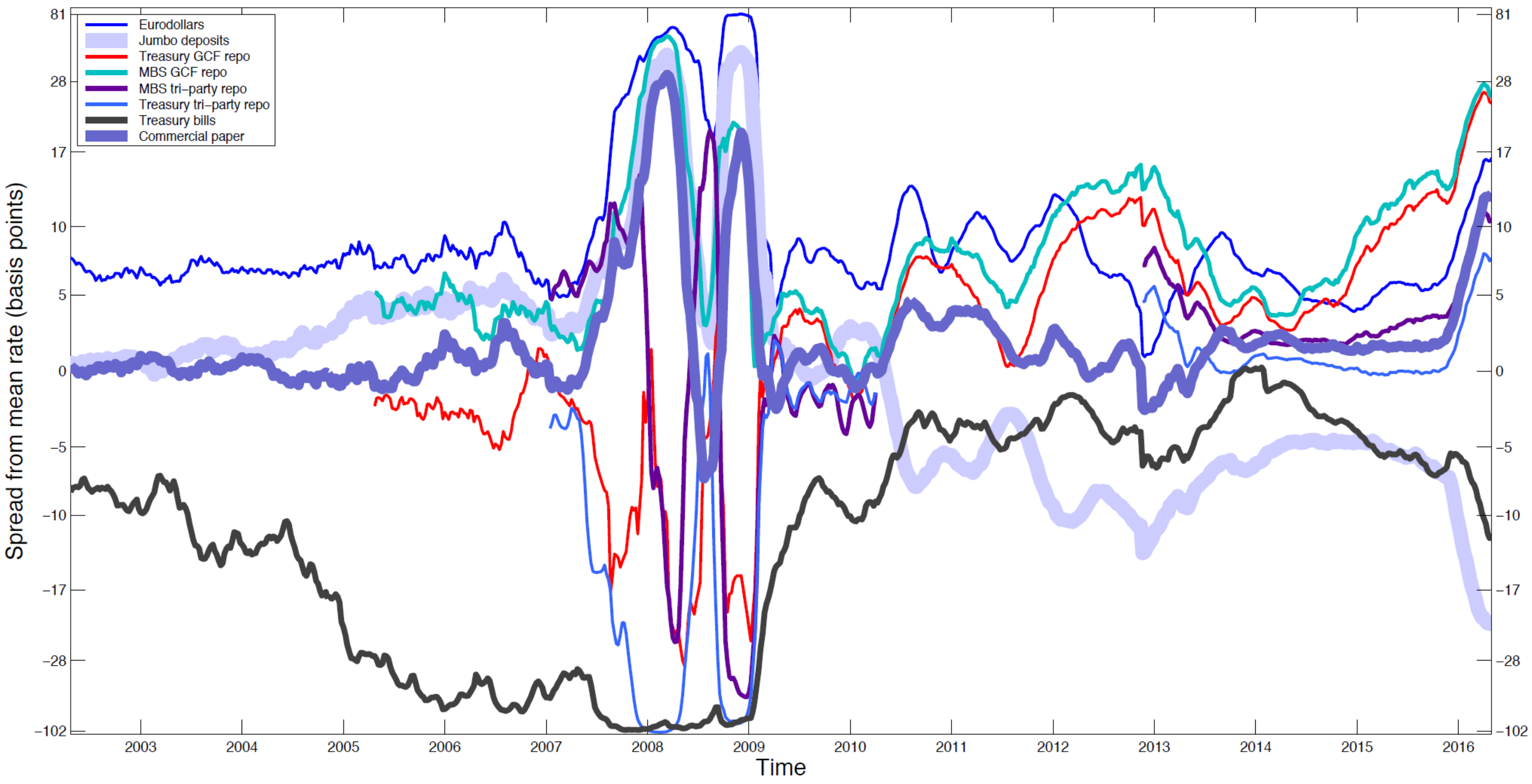
- Repo: secured by Treasury and Agency collateral
- Large, safe, money market

For financial intermediaries:

- Bank payment and settlement needs: Repo ($trillions) has largely replaced the federal funds market (<$100 billion)
- Hedge funds/dealers and cash-futures arbitrage (Barth and Kahn, 2025; Kashyap et al. 2025)

Policy rate passthrough

Policy rate passthrough across money markets is imperfect



Rate dispersion from Duffie-Krishnamurthy (2016), Jackson Hole Symposium

New money markets and the Fed

Fed now has a significant native presence in the repo market

- Reverse repo deposit facility (2013)
- Standing repo facility (2021)
- *Expanded beyond the traditional banking perimeter*

LESSON:

Fed needs a native presence in the central money market

- Ensure smooth functioning of payment system and securities market settlement
- Maintain “no-questions asked” equilibrium

Another money market?

24-7 stablecoin payment platforms, instant settlement, smart contracts

Reverse Gresham’s Law: “better money drives out worse”

What will the Fed’s toolkit look like?

- Stablecoin deposit and repo facility
- Asset/liability regulation

Congress has passed a four-year ban on a Fed CBDC

- How fast will technology evolve?

Assets	Liabilities
Treasurys	Reserves
Standing repo facility	Reverse repo
	Digital coin?

International money is a liquidity tournament

The reserve currency has historically been housed in the currency with the largest and most liquid safe asset market

Dutch Florin (17th/18th cent.) → Pound Sterling → U.S. Dollar

- Eichengreen (2005), Quinn and Roberds (2019), Coppola, Krishnamurthy and Xu (2026)

The US has led and still leads other economies in terms of market size of safe and liquid assets:

- Supply: Treasury + Repo + Banking ~ $20 to $30 trillion

Developments have widened the lead:

- Repo technology in US
- European sovereign debt crises

Will new technologies increase or decrease dollar demand?

Dollar stablecoins provide cheaper access to dollars, boosting demand for dollar assets

- Coppola, Krishnamurthy, Xu (2026); Liao, Prasad, Zhang (2026)
- Reducing Treasury yields (Ahmed and Aldasaro, 2025)

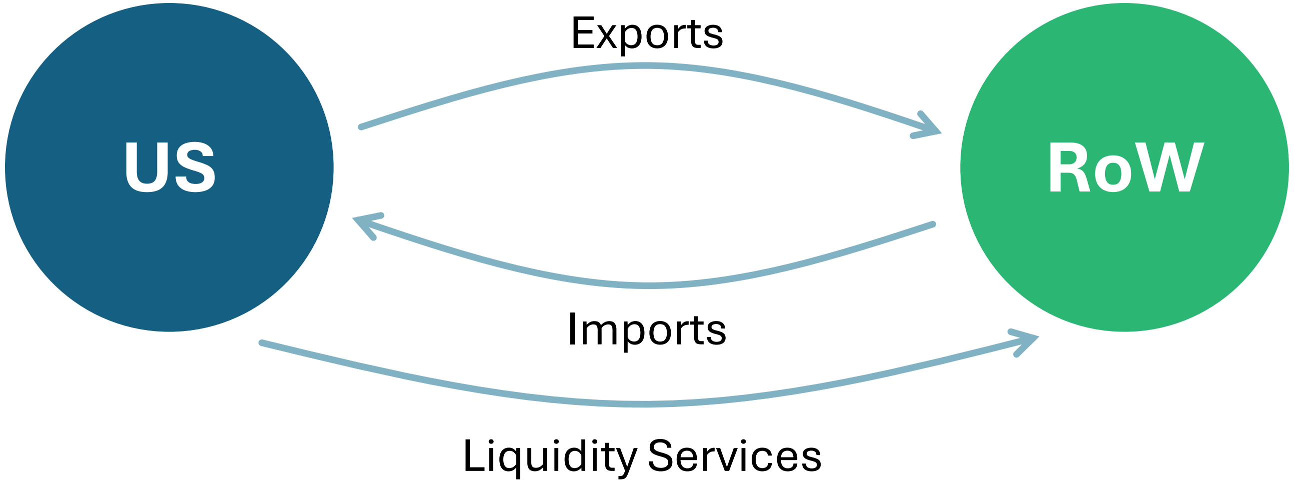
Tokenization of securities effectively expands supply of safe assets

- Which reinforces dollar demand via complementarities
- CKX (2026)

What if all currencies adopt better technologies?

Ambiguous: CKX theory and history do not offer clear guidance

CKX: More at stake for the dominant currency



"Liquidity exports" = Quantity × Convenience yield

Quantity of foreign-held safe assets

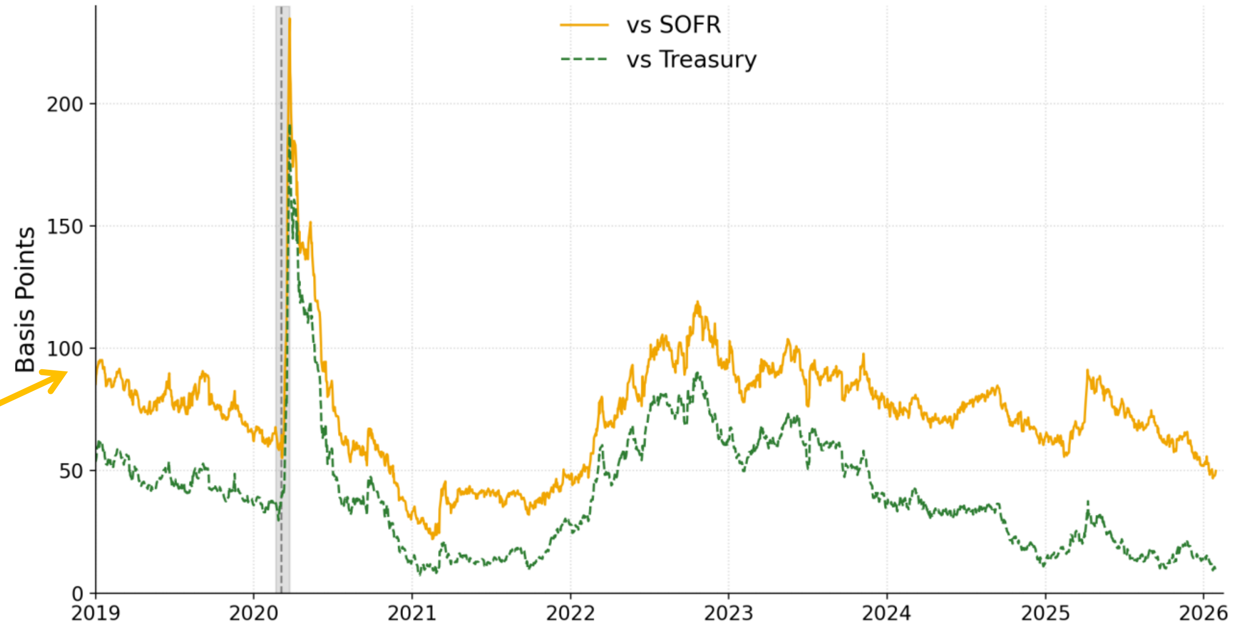
Convenience yield on safe dollar assets

How large is the convenience yield on safe/ liquid dollar assets?

- Krishnamurthy and Vissing-Jorgensen (2012): 75 basis points on US Treasury versus US corporate, 1920-2012

- Repo is the new dollar safe asset

- Jiang, Krishnamurthy, Lustig (2021): 200 basis points
  - Treasury relative to non-dollar assets



Orange = 10-year corporate bonds with a CDS hedge minus 10-year repo (SOFR) swap

Estimate of what is at stake

- Using the 200 bps estimate + estimate of foreign holdings of dollar safe assets
- Seigniorage is ~1% of GDP
- Present value of seigniorage is ~100% of GDP, or ~$33 trillion
  - Accrues to taxpayers (via government budget), banks, mortgage finance

See Jiang, Krishnamurthy, Lustig and Richmond (2026)

Conclusion

Suppose financial innovation leads to better financial technology

Replaces existing money with new money

- Even if it's only marginally better: Reverse Gresham's Law "better money drives out worse" leading to migration of money
- Financial stability concerns: Fed needs a native presence

If technological improvements are significant

- Outcomes in the international money tournament change
- Design and regulation will factor in selecting the winner

A lot at stake, and getting the regulatory and Fed toolkit right matters